Greening Your Home: The Foundation

JIM LARUE

Jim LaRue's company is The HouseMender, Incorporated. He has prepared a "Greening Your Home" series for the Cleveland Green Building Coalition.

Whether you are building homes using foundations made of concrete block, poured concrete, concrete poured into foam insulation forms, precast concrete walls, or some other system, greening these systems means paying attention to several matters that reduce any negative impact that the foundations could have on the environment and the people living in the home.

If the site is large enough, we want to keep most of the topsoil on the site. The topsoil, unless it is laced with lead or other contaminants, or is from a brownfield, is the natural soil for this site. Removing it and then replacing it not only adds an unnecessary transportation (and therefore, environmental) cost, but also spoils the natural conditions that could support plant life common to the area. We also want to use the smallest foundation necessary, so that displacement of large amounts of soil is not required. If a slab-on-grade home is to be considered, green building methods would recommend the use of a frost-protected shallow foundation, which can usually be constructed without excavating more than 1 ft below grade. Builders using this system have significantly reduced their building costs.

Poured-on-site concrete walls are being used by many builders, and there are several ways to make these foundations greener:

• Fly-ash, a by-product of coal combustion, can be used as a substitute for cement; currently as much as 20% replacement is recommended. Using this waste product for this purpose is a significant recycling measure.

• Using recycled concrete or glass cullet for aggregate in the concrete is yet another way to recycle waste products and reduce the use of mined aggregate.

• Using reusable aluminum forms instead of wood forms means reducing wood waste.

• Using low-toxicity release agents on the metal forms is important. On my first green home, I discovered that the crew was using used motor oil to lubricate the forms. I ended up scrubbing interior wall surfaces to remove oil refuse from the surface. Special release agents are now available which are environmentally friendly (see, for example, http://concrete-solutions.com/ and www.wrmeadows.com/wrmustoc.html/).

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Foam insulation is now being used with foundations in a variety of ways. The good news is that the foam is effectively working to reduce heat loss through the foundation walls. The bad news is that termites have discovered that they can work year-round on the wood in a home if the foam insulation on the exterior can insulate them from the cold surface soils. Some of the foams now incorporate some borates that help reduce this infestation a bit, but borates can lose their effectiveness if they are exposed to moisture for too long and too intensely.

One attractive feature of the precast foundation is that the foam insulation is placed on the interior side of the concrete walls where the termite infestation is not likely to occur. This also means that instead of foam insulation on the exterior to the soil level and another kind of insulation on the interior above grade, there is a common plane of insulation from floor to ceiling on the inside. This makes for a much more effective insulation plane. Another positive feature of the precast units is the concrete ribs on the interior, which have spaces between them that can be filled with additional insulation (as much as R-19). A furring strip secured to the front edge of each rib makes it easy to install a surface finish material.

Another use of foam in a foundation system is the placement of 2 inches of expanded foam under the basement slab. (Use of subslab foam insulation in warm climates, where most homes are built on slab-on-grade, is probably not appropriate.) This measure means that room heat is not lost through the floor, and the floor feels more comfortable because it is storing the heat generated in the living space.

I have investigated numerous problems in basements where moisture in the basement has condensed on the cold floors and caused mold problems under and in the carpeting placed right on a concrete floor. I generally recommend against placing carpeting on a basement floor (my mantra is there is no such thing as a permanently dry basement), but if the concrete is insulated, it is less likely to experience a dew point temperature that can be a problem if room humidity levels are high.

Speaking of moisture, green building is, of course, greatly concerned with keeping moisture out of the basement interior. Most foundation systems involve applying a dampproofing material on the exterior of the walls below grade. Using a nonsolvent-based dampproofing, so there is no leaching of these

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materials into the soil around the home, is strongly recommended. Leaching of these materials into drain tile can result in its eventual appearance in treatment plants, streams, and rivers. It can also contribute to soil gases that can enter the living space.

Precast foundations, which are brought to the site in pieces and are placed in very little time, provide another effective measure; their greater strength and reduced permeability minimizes any wicking of moisture through the walls and dampproofing is not necessary in the first place. Installing a nonsolvent dampproofing or even some 6-mil poly on the footer before the walls are poured or placed is another way to reduce penetration of moisture wicking up through the footer, where it can enter the basement area.


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Precast foundations, which are brought to the site in pieces and are placed in very little time, often provide some green advantages over poured-on-site concrete walls.

Radon is another soil gas that can enter the living space. Radon can occur in soils almost anywhere, including in the water table beneath our homes, and can be found in any home if the conditions are right. Green building methods strongly recommend installing a subslab vent to remove radon, moisture vapor, and other soil gases via a 4 inch pipe from the slab up through the roof of the house. Consider installing an inline fan in this pipe if passive venting is not sufficient. These methods provide cheap insurance that helps avoid an expensive retrofit later if soil gas problems surface.

Using crushed concrete, rubble, or even rubber tire scraps, if they are available, as backfill around a foundation is another way to use recycled materials. There is a growing supply of these materials, so transportation costs are less of a concern than they once were. Wrapping them in a geotextile helps to prevent soil fines from getting into the backfill and blocking surface water from reaching drain tile.

Green building methods strive to ensure that every foundation is as moisture protected (to reduce mold potential and structural decay), energy efficient, and environmentally sensitive in its use of materials as possible. Foundation systems, such as the ones I've discussed, are emerging that can effectively accomplish these objectives with a minimum amount of maintenance for the life of the structure. Indeed, the life of the structure is significantly increased when these measures are considered and included.